10-min knowledge sharing/discussion

	12313505	胡伟毅
4	**12213005**	**洪锦奕**
	12310429	**方一州**
	12212216	**孙佳琦**
5	**12210924**	**张宇轩**
	12311626	**郑金辉**
	12212317	**孟嘉旭**
6	**12210452**	**唐易凤**
	12211725	**郭家麟**
	12312731	雷灿
7	12211122	邓卜凡
	12313636	石毅东
	12310807	姜力维
8	12210930	王之一
	12313609	杨竣杰

12211024	唐稚量	
12210121	文绎博	9
12212210	李钇漳	
12312633	曾紫涵	
12112725	王宇	11
12211127	钱伟烨	
12310108	张洪源	
12211623	张仁睿	12
12310403	安文博	
12212454	刘夕媛	
12310906	邱彦鸣	13
12313415	唐梓朕	
12212120	万文博	14

Speech Signal Processing

LAB V
Frequency-Domain Representations

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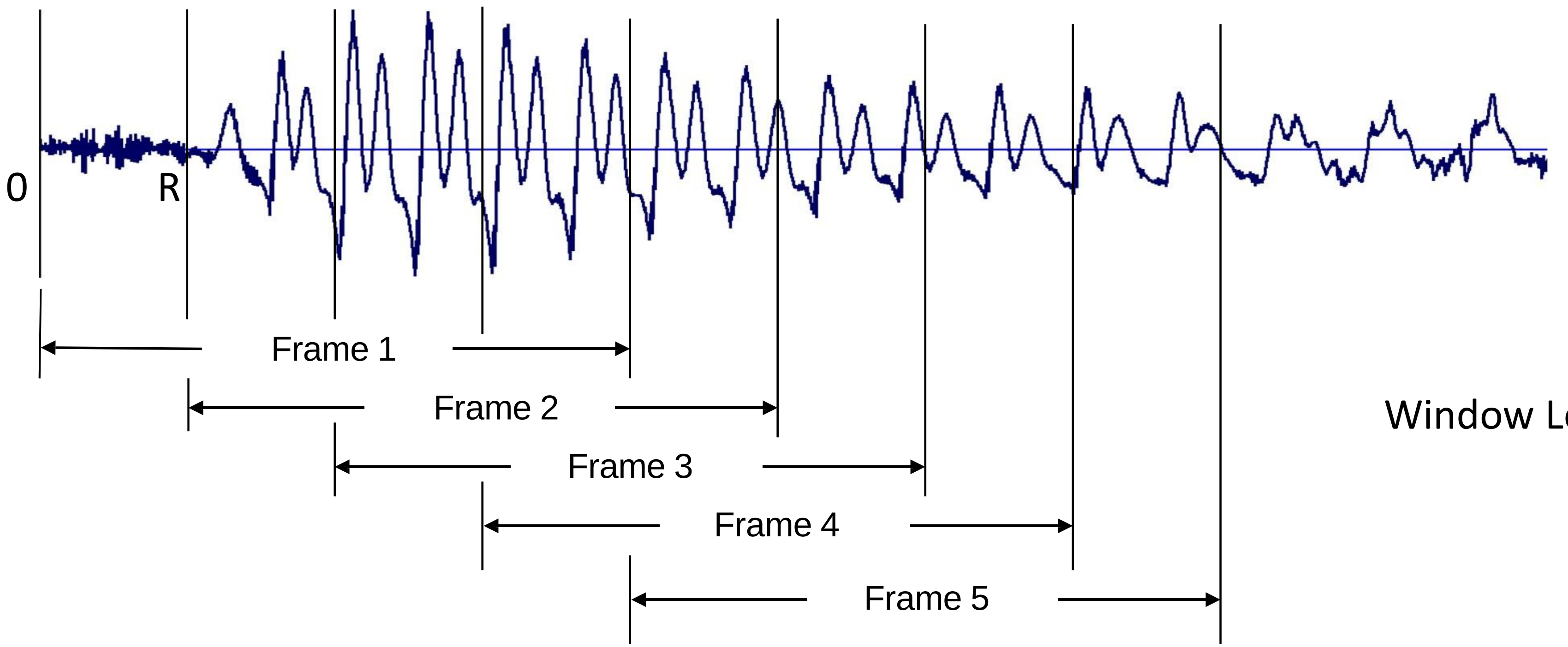
Purpose of this lab...

1. Learn to analysing speech signal with short-time frequency analysis/spectrogram
2. Learn the windowing effect for short-time frequency analysis

Short Time Processing



Window Length: L

Frame1={x[0],x[1],…,x[L-1]}
Frame2={x[R],x[R+1],…,x[R+L-1]}
Frame3={x[2R],x[2R+1],…,x[2R+L-1]}
…

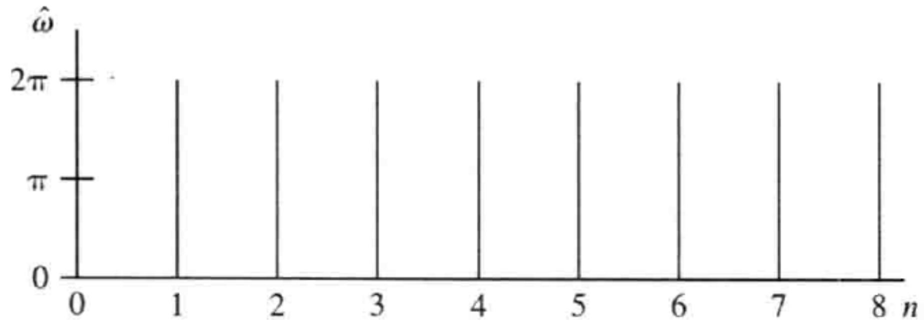
Short-Time Fourier Transform

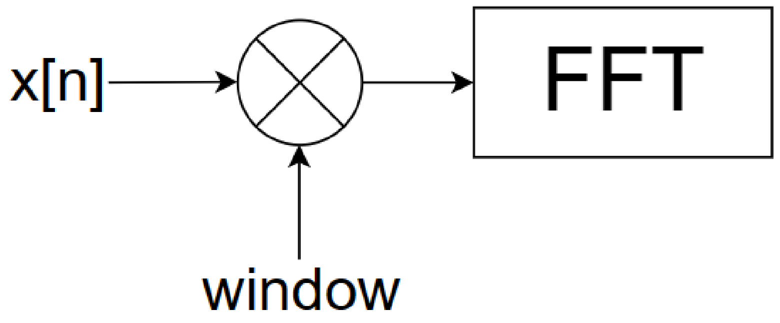
$$X_{\hat{n}}(e^{j\hat{\omega}}) = \sum_m w[\hat{n} - m]x[m]e^{-j\hat{\omega}m}$$

- when $\hat{n} = \hat{n}_0$

$$X_{\hat{n}_0}(e^{j\hat{\omega}}) = \sum_m (w[\hat{n}_0 - m]x[m])e^{-j\hat{\omega}m}$$

$$X_{\hat{n}_0}(e^{j\hat{\omega}}) = DTFT\{w[\hat{n}_0 - m]x[m]\}$$





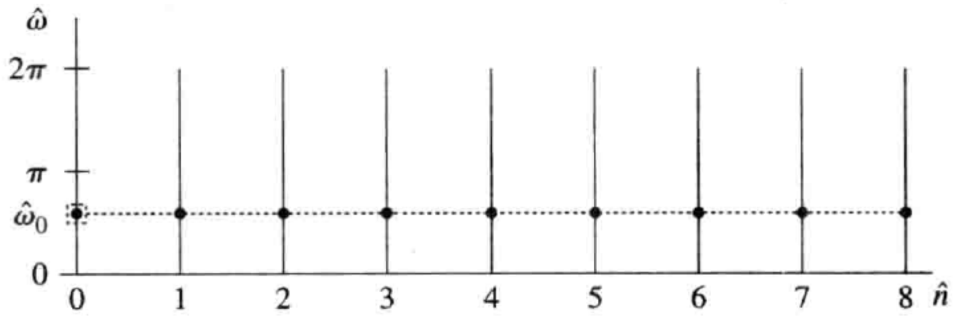
Short-Time Fourier Transform

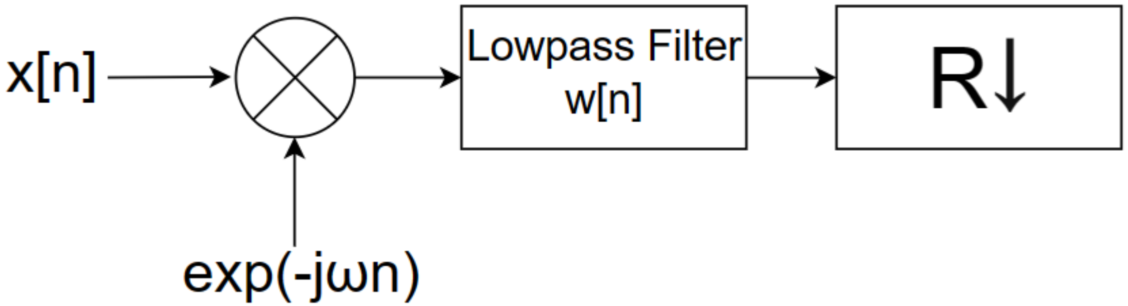
$$X_{\hat{n}}(e^{j\widehat{\omega}}) = \sum_m w[\hat{n} - m]x[m]e^{-j\widehat{\omega}m}$$

- when $\widehat{\omega} = \widehat{\omega}_0$

$$X_{\hat{n}}(e^{j\widehat{\omega}_0}) = \sum_m w[\hat{n} - m]\left(x[m]e^{-j\widehat{\omega}_0 m}\right)$$

$$X_{\hat{n}}\left(e^{j\widehat{\omega}_0}\right) = x[m]e^{-j\widehat{\omega}_0 m} * w[m]$$





Problem 1

7.24. (MATLAB Exercise) Write a MATLAB program to perform short-time spectral analysis on a single frame of the speech waveform. Your program should accept the following inputs:

STSA_SingleFrame (filename, startsmp, framelength)

1. the speech filename
2. the starting sample within the speech array for frame analysis
3. the frame length in msec

You will have to convert from the input frame length in msec to the frame length in samples, based on the sampling rate of the digital speech signal being analyzed. Using a Hamming window of the appropriate length, window the selected frame of speech and plot (on a single page) the following signals:

1. the speech waveform for the entire duration of the speech file
2. the windowed speech frame beginning at the desired starting sample
3. the magnitude of the STFT of the selected frame of speech

zero padding

4. the log magnitude (in dB) of the short-time Fourier transform of the selected frame of speech

Use the following input to test your code:

- speech files: `s5.wav` and `vowel_iy_100hz.wav`
- starting samples: 7000 (for file `s5.wav`) and 1000 (for file `vowel_iy_100hz.wav`)
- frame length: 40 msec for both examples

Problem 2

7.25. (MATLAB Exercise) Write a MATLAB program to perform (and compare the results of) multiple short-time analyses of a section of speech using multiple window lengths. Your program should accept the following inputs:

1. the number of short-time spectral analyses to be compared
2. the speech filename
3. the starting sample within the speech array for analysis of the short-time spectrum
4. the frame lengths in msec for each of the multiple analyses

STSA_MultiLengths (num, filename, startsmp, framelengths)

vector

For each of the spectral analyses you will have to convert from the input frame length in msec to the frame length in samples, based on the sampling rate of the sampled speech signal being analyzed. You should do the short-time analyses using both a Hamming window and a rectangular window so as to be able to contrast the results for these two windows. Your program output should be a plot with the multiple spectral analyses when using a Hamming window, and a separate plot with the multiple spectral analyses using a rectangular window. Each plot should include four sub-plots with the following:

1. the speech waveform for the file being analyzed
2. the waveforms of the speech segments being analyzed, showing the window superimposed over the waveform
3. the magnitude of the short-time transforms, using a different color for each of the analyses
4. the log magnitude (in dB) of the short-time transforms, again using a different color for each of the analyses

Use the following input to test your code:

- speech file: `s5.wav`
- number of analyses: 4
- frame lengths of analyses: 5, 10, 20, 40 msec
- starting sample for analysis: 7000

Problem 3

7.26. (MATLAB Exercise) Write a MATLAB program to create digital spectrograms (both narrowband and wideband spectrograms) from a specified speech file. For this exercise you can either use the MATLAB routine `spectrogram` or write your own code to* create the required sequence of short-time spectra for plotting as a spectrogram.

*bonus: write your own code

Your program should accept the following parameters for creating a wide range of spectrogram plots:

- the speech filename
- an option to lower the sampling rate from the original rate of the speech to a lower rate, e.g., from 16 kHz to 8 kHz
- the window length (in msec) for the wideband spectrogram
- the window length (in msec) for the narrowband spectrogram

$\Big\}$ [Lwide, Lnarrow]

- the FFT length for the wideband spectrogram computation
- the FFT length for the narrowband spectrogram computation

$\Big\}$ [FFTwide, FFTnarrow]

- an option to plot either the log magnitude or the linear magnitude of the short-time spectra imagesc (x, y, C) or pcolor (x, y, C)
- the dynamic range of the spectrogram image clim (limits) or caxis (limits)
- an option to display the spectrogram in color or gray scale colormap ('gray')

Your program should plot both the narrowband and wideband spectrograms on a single page.

STSA_Spectrogram (filename, resamplerate, windowlengths, FFTlengths, magscale, range, color)

```
h = pcolor(t, f, mag2db(abs(s)));
set(h, 'LineStyle','none');
```

or

```
imagesc(t, f, mag2db(abs(s)));
set(gca,'YDir','normal');
```

Test your program with the following inputs:

- speech filename: `s5.wav`
- resample option: 0 (don't change sampling rate)
- window length for wideband spectrogram: 5 msec
- window length for narrowband spectrogram: 50 msec
- FFT size for wideband spectrogram: 1024
- FFT size for narrowband spectrogram: 1024
- option to use log or linear magnitude: use log magnitude
- dynamic range of spectrogram image: 60 dB
- color option: plot in gray scale and in color (two separate plots)

Once you get your program working well, experiment with utterances at different sampling rates, and vary the analysis parameters to see their effects on the resulting spectrograms.